User administration

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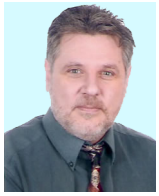

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Introduction

This module focuses on the user and everything related to the following:

- Servers
- Workstations
- Software
- Data
- Incidents
- Training

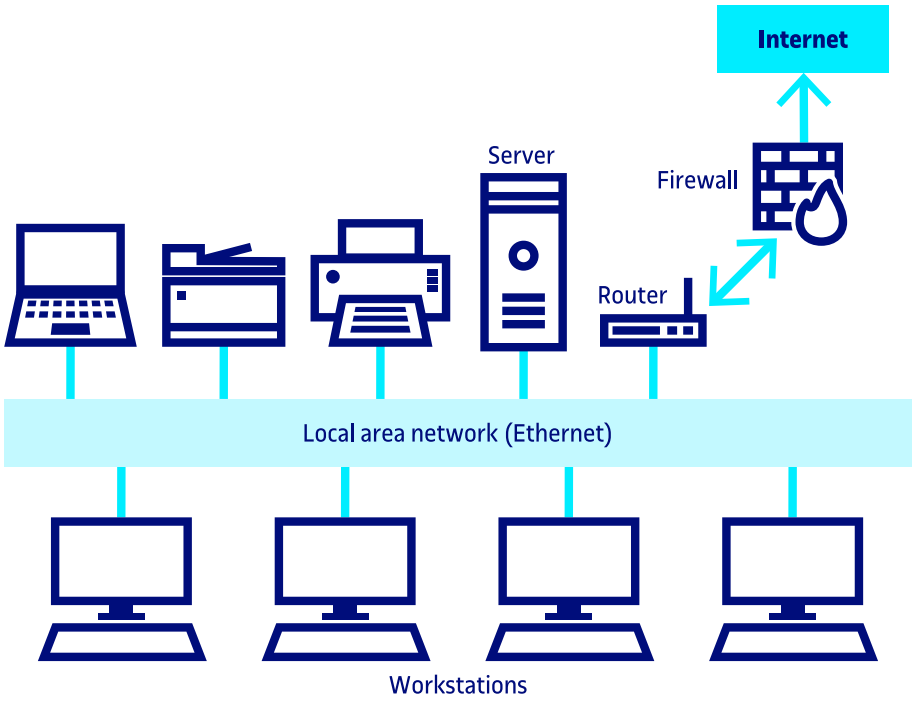
The function of IT systems is to provide users with support and services, and to help them do their jobs within the organisation.

The department's philosophy

Some authors tend to view IT services as a business and users as customers. When this is the case, the IT department's philosophy is based on organisation, work and performance at all times.

Figure 1 depicts a general outline of all the networks. As shown, there are two distinguishing elements: workstations and servers.

Figure 1. General structure of a network



Therefore, from an administrative perspective, how users are handled can be divided into services or functions that must be performed on the server side and services or functions that must be performed on the user or client side.

First, you will see how the user environment is designed from the servers perspective. A good design makes administration easier and provides users with a more consistent and robust environment. In the end, this also makes it easier for them to use.

Next, you will learn about workstation installation, configuration and maintenance. A well planified configuration and a mechanism for recovering configurations can provide good troubleshooting services and, once again, simplify workstation administration. This covers the part that concerns the user environment.

The following sections deal with training and user incident or issue support. These are just as important as the previous ones, but are often neglected. Training reduces costs in the medium term, saving IT department time and making organisation staff self-sufficient in many tasks. Proper incident management[1] also cuts costs for the organisation and, indirectly, for the IT department.

(1)This is often called a *help desk*.

Objectives

The learning resources in this module cover the content and key tools needed to achieve the following objectives:

1. To learn how to design a suitable user environment for the organisation.
2. To learn how to differentiate between tasks that affect servers and those that affect workstations.
3. To learn how to design a useful workstation environment for users that can be managed as easily as possible.
4. To learn how to streamline the IT department so that it can respond quickly to user incidents.
5. To learn how tasks can be systematised into procedures.
6. To understand the responsibilities of the user administrator.
7. To know tools for configuring, maintaining and recovering workstations in problematic situations.

1. User environment design

Hardware-wise, the IT system includes servers, the network and workstations. Software-wise, it includes operating systems and applications. Putting them all together is not enough. They must be designed to interact with each other to achieve the desired result.

When defining what the user will see when connecting to the organisation's servers, the user environment is designed, which impacts everything from aesthetics to the available resources and applications.

Designing a **user environment** involves preparing everything that users will find when using the organisation's IT systems.

The following criteria and objectives should be considered when designing:[2]

(2)The list is a rough guide.

Reflection

Do you think your organisation's environment satisfies all these objectives you consider important that are missing? Leave a comment in the course forum.

- It should be simple to use and intuitive for the user.
- It should give all users a consistent environment.
- If computers or workstations change, the environment (software and hardware) should not feel unfamiliar.
- The system should be fast, with quick server response times and fast network speeds.
- It should provide a good level of security.
- It should be easy to administer.
- The software should be easy to update.
- If the computer crashes, it should be easy to reinstall.
- If the computer loses its configuration, it should be easy to reconfigure.
- If the computer crashes, no information should be lost (as little as possible, and it should not be critical).
- It should be easy to make backups.

- It should be easy to respond to a workstation disaster.

In an ideal world

Given this list of objectives and design criteria, the first thing that seems obvious is to try to achieve the following:

- The entire software environment should have a homogeneous interface.
- The entire workstation hardware environment should be homogeneous.

Although these are two very interesting objectives, they are difficult to achieve in practical terms. However, it is more feasible to consider making the user and hardware environments as homogeneous as possible.

Considering these objectives, the number of existing corporate servers, the existing network, the workstations installed, knowledge about the organisation's workstations, the structure of the departments and the organisation, etc., the environment in which users will work many hours a day must be designed. Careful planning is therefore essential.

Contradictions

Design criteria and objectives tend to contradict each other. Security is often at odds with convenience and speed. The end result is always a trade-off between these elements.

The user environment design affects both servers and workstations and must therefore be approached with both sides in mind (as a single unit). They work in tandem, so it is not possible to design one part without considering the other. Once the principles design has been established, each part can be further described.

1.1. General user requirements

Every organisation is different. However, IT user requirements can be generalised to some extent. Generally speaking, all users have the following requirements when it comes to IT systems:

1) A workstation. This is generally a computer. Some users may need special devices such as DVD drives, scanners, local printers, label printers, etc.

2) A place to print. This can be a local printer or a remote printer shared with other users. Companies usually have a shared printer connected to the network in each department, where users can send their documents, instead of each user having their own local printer.

3) Space to store information. This can be done locally on the user's workstation hard drive or remotely on servers.

4) Software for work. All the programs and applications that users will need to work.

Generally, the user's software requirements can be divided into several categories:

- **Base software**: operating system and basic communications applications on servers.

- **Office software**: office suites, which usually include a spreadsheet, a word processor, a database and a calendar.

- **Communications software**: this generally includes email, an internet browser (Edge, Firefox, Chrome, etc.) and software for video calls (Teams, Google Meet, Zoom, etc.).

- **Specific applications**: this is the group of applications that are specific to the organisation or even the department, such as invoicing, accounting, graphic design, production control and payroll programs, etc.

Reflection

Do you think any software categories are missing? Leave a comment in the course forum.

When accessing most specific applications, users are typically asked to enter a username and password (in addition to those used to log in to the network). This login information, which is usually required for applications with databases on servers, is used to assign user privileges within the application, meaning that the client side is identical for all users and what can be done depends solely on the user who logs in.

Single sign-on

Users want to log in once and access all systems. There are several mechanisms for this, such as:

- SAML
- OAuth 2.0
- OpenID

Access to these applications (both standard and specific) must be controlled somehow so that not everyone has access to the entire organisation's data. To handle the organisation's data (i.e., create, modify or view it) the user does not need to know where the data is located, only how to access it and how to use it for their work.

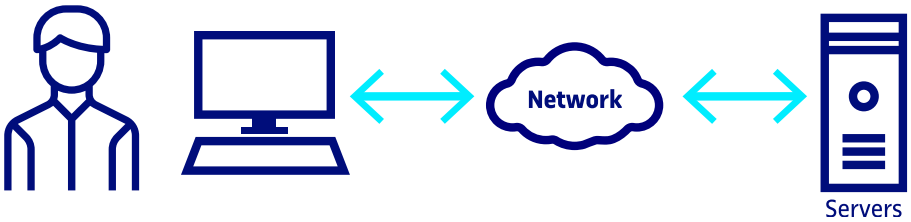
For everything to work properly, all these features need to be in a user-friendly environment easy to use. If the system doesn't help with the task, it just makes it harder, and instead of improving performance, it actually makes it worse and slows down the flow of information throughout the organisation.

Although users have very different roles in the company depending on their position, all users in an organisation can be put together in similar **requirement groups**. There will not be many groups, nor will they be very different from one another.

1.2. The IT system and the user

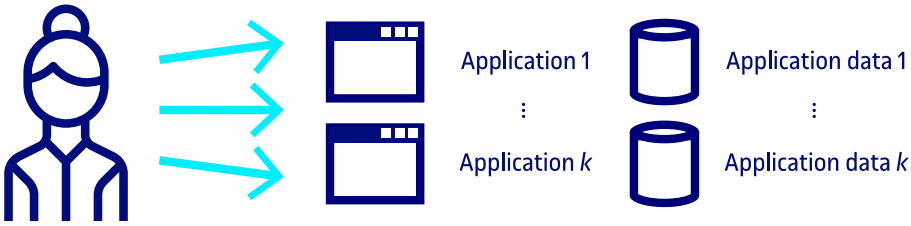
The idea a user might have of an IT system is similar to the one shown in Figure 2.

Figure 2. IT system diagram



Meanwhile, the idea a user might have of applications is similar to the one shown in Figure 3.

Figure 3. Application diagram



However, users do not know where applications are physically installed or where data actually is stored within the organisation's IT network.

IT mysteries

Users often find it astonishing when they go to another computer, log in and find all their data and applications. Similarly, if something has been saved (locally), they do not understand why it is not there when they go to another computer, since they "saved the same as always", and they show many files as proof, while the one they saved an hour ago has "vanished". People do not learn everything overnight, and this is perfectly normal. Users are gradually being educated in these new work tools.

Based on the above list of objectives and knowing how users view the IT network, different access control mechanisms will be examined to enable the user environment to be designed.

1.3. Access control

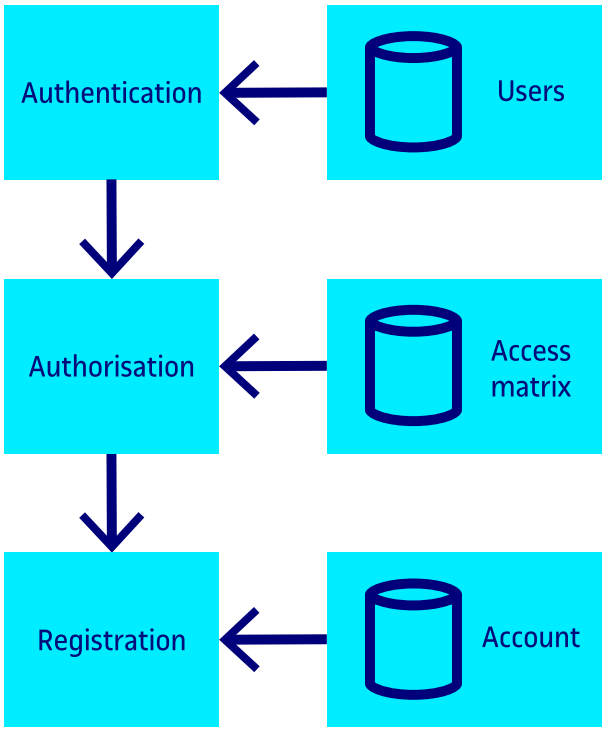
A key design consideration for user environments is ensuring that users can only access what they need. To access a resource on the IT system, users must identify themselves (**authenticate**). Once identified, the system controls (**authorises**) access to the IT system's resources and records (**account**) how each resource is used,[3] as shown in Figure 4.

(3)This is known as the AAA security model (authentication, authorisation and accounting).

Note

Access control is an extensive security topic that goes beyond the scope of these resources. Here, only that part which is useful for computer system design will be discussed.

Figure 4. Diagram of the AAA security model



Principle of least privilege

Grant the most restrictive set of privileges (the lowest authorisation) necessary to carry out the task.

Source: **Department of Defense**. "Department of Defense Trusted Computer System Evaluation Criteria Document". *Orange Book* (DOD-5200.28-STD).

The process of **authentication** refers to verifying the identity of a person or process that intends to access the resources on an IT system. This is typically accomplished via a username and password or token.

See also

For more information on authentication, see the *Security administration* module.

Authorisation refers to the process whereby the system allows the identified user to access the resources of an IT system.

Authorisation determines what access is allowed to each entity. Authentication verifies a person's identity, while authorisation verifies that a known person has the authority to perform a certain operation. Authentication must therefore precede authorisation.

Access control determines a user's privileges on the IT system and which resources they can access. This access control must be carefully supervised, as it could otherwise lead to problems accessing private information belonging to other individuals or groups within the company.

Finally, **registration** logs the user activity on the IT system.

1.3.1. Access control matrix

The access control matrix or access matrix is a formal model of IT security used in IT systems, which describes the rights of each subject regarding all objects in the system. Objects are entities that contain information and may be physical or abstract. Subjects access objects and may be users, processes, programs, or other entities.

The most common access rights are: read access (**R**), write access (**W**), and execute access (**X**).

Table 1 shows a typical access matrix. Rows in the matrix represent domains (or subjects) and columns represent objects. Entries in the matrix consist of a series of access rights. For example, the entry $access_{(i,j)}$ defines the set of operations that a process, running in domain D_i, can perform on an object O_j.

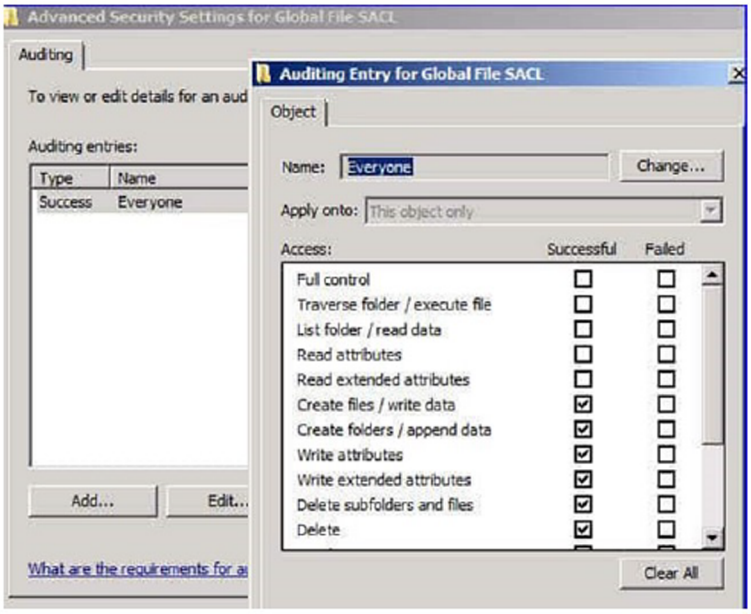
Table 1. Access rights matrix example

Domain	Object	
	File	Directory
D1	Read	Read Write Execute
D2		Read Write
D3	Execute	Read

1.3.2. Access control list (ACL)

The matrix is not usually saved because it is too large. Most domains do not have access to most objects, so storing a huge, almost empty matrix would be a waste of disk space. Instead, each object is associated with an (ordered) list of all the domains that can access it and the way in which they can do so. This list is called an **access control list** (ACL), and an example is shown in Figure 5.

Figure 5. Access control list



1.3.3. Single sign-on

This is a term used to identify single login or unified login. Single sign-on (SSO) refers to an authentication procedure that grants a specific user access to multiple systems within an organisation using a single instance of identification. Once a device has been authenticated, no further authentication is required to access email or instant messaging.

This system has its pros and cons. An advantatge is that it avoids users from having to log in to different apps multiple times. However, a drawback is that if someone logs in to a user's workstation (or the user leaves the session open), they can access all the apps that user is authorised to use without having to validate their identity again.

1.4. IT system design

The next subsections analyse how to apply the access matrix in different designs and study its advantages and disadvantages.

1.4.1. Minimum security

This design simplifies administration considerably, based on the following **criteria**:

Reflection

How does this solution sound? You can share your thoughts in the course forum.

- All users can see all programs and applications.
- All users have minimum permissions (read and execute) for everything.
- Specific applications (such as databases), which already have their specific access permissions, are controlled by the application itself. No special permissions are required.
- User folders can only be accessed by the same user with read, write and execute permissions.

The **advantages** of this design are as follows:

1) It simplifies administration, since all users are considered to have the same rights, and therefore creating a user does not require any special action.

2) It makes it easier to set up the base image for workstations.

3) Once a computer has been cloned, few final adjustments are needed.

4) Any user can run any application, so modifying the organisation's workflows does not pose any problems or require any changes to the existing IT structures.

5) Changing workstations does not involve the added cost of installing specific applications for that user, as all applications are available on all workstations.

6) The working environment is completely uniform across the entire organisation, as everyone has access to the same applications and resources.

7) It makes it easier to search for information within the IT system because all information is "flat" (not hierarchical or protected) within the system.

8) Users have access to the full IT potential of the organisation.

9) Any change involving the use of new software does not require modifications to the IT system.

The **drawbacks** of this design are as follows:

1) The idea of a work group, group of people, department, etc., ultimately, the grouping is not included in the IT structure, which can complicate administration when it needs to be incorporated into information management.

See also

See the subsection "Creation of the base image for workstations".

2) Sharing information between user groups is not easy, since there is no concept of a group of people. For example, it is not possible for an entire department to share information itself without the rest of the organisation having access to it.

3) Users may become confused by the huge amount of available software, as they may not know which they should use for work and which not.

4) Sharing a user's directory to people in their workgroup, if desired, is not possible. Doing so would automatically share to the entire organisation.

5) It may be problematic to allow anyone in the organisation to run any application. There may be sensitive information that should not be accessible to other groups within the organisation.

6) Many requests from user groups regarding the handling of information, especially if it is sensitive, are too complex or even impossible to fulfil.

7) There is a risk of mishandling information, leading to unwanted outcomes.

1.4.2. Users in groups

Another solution is possible by modifying certain criteria in the minimum security design. This solution is more secure than the previous one and is based on the following **criteria**:

Reflection

How does this solution sound? You can share your thoughts in the course forum.

- Users naturally form groups within the organisation, and these groups are reflected in some way in the IT system.
- A user can only belong to one group.
- An application can work for everyone or only for a group.
- All users have minimum permissions (read and execute) for the group objects.
- All users have read and execute permissions for general objects (everyone's).
- Specific applications (such as databases), which already have their specific access permissions, are controlled by the application itself. No special permissions are required. Now only user groups that need these applications can see them.
- User folders can only be accessed by the same user with read, write and execute permissions.
- The software categories that could be found within an organisation must be considered.

The **advantages** of this design are as follows:

1) All users are equal within groups, so creating a user means considering which group they should be part of.

2) Any user is only authorised to run any application in the group and are common to everyone.

3) Users can only access group information and common information. This means that the organisation's information is considerably better protected.

4) Incorrect software handling is impossible, as only users in the group can now run it.

5) The working environment is fairly consistent across the organisation, but varies depending on the applications allowed to be used in the group. Moreover, the application suite common to everyone is the same, which gives the user a strong sense of consistency.

6) Users essentially have the software resources they need. It makes it easier for them to know which software is available to them and which they should use, unlike before, when they would see software they were not supposed to use.

7) The idea of a work group is now embedded and is very useful for sharing information within the group and for working on specific applications in a coordinated manner.

8) Now, a user's directory can be shared to people in their work group, if desired. If this is done, it is automatically shared only to their workgroup.

See also

See the subsection "General user requirements".

The **drawbacks** of this design are as follows:

- A change in the workflow may mean modifying all permissions for applications and work folders for groups, i.e., having to modify the existing IT structures.
- If someone moves to a different group, their profile will need to be updated because they will have access to new applications and will lose access to some of the ones they had before (those specific to their group).
- Searching for information within the system is more complex, as it is now organised by work groups within the organisation.
- Installing new software can be a serious problem if it is to be used by several work groups.
- Sharing information between groups is complex.

1.4.3. Users in multiple groups

This solution is an extension of the previous one (users in groups) because some users belong to more than one group. Groups work well in most cases, but for some they are not sufficient. For example, if the organisation uses work groups within the department, or if subgroups are created within the work group, a person may belong to more than one group at the same time. Therefore, this should be analysed by first assessing the **criteria** to be followed:

Reflection

How does this solution sound? You can share your thoughts in the course forum.

- Users join groups naturally.

- A user can belong to one or more groups.
- An application can work for all or for one or more groups.
- All users have minimum permissions (read and execute) for the group objects.
- All users have read and execute permissions for general objects (everyone's).
- Specific applications (such as databases), which already have their specific access permissions, are controlled by the application itself. No special permissions are required. Now only user groups that need these applications can see them.
- User folders can only be accessed by the same user with read, write and execute permissions.

The **advantages** of this design are as follows:

1) All users are equal within groups, so registering a user means considering which groups they should be part of.

2) Any user is only authorised to run any application in their groups and all applications that are common to everyone. They also have permission to access common group information. A change in the workflow may mean modifying all permissions for applications and work folders for groups, i.e., having to modify the existing IT structures.

3) If someone moves to a different group, their profile will need to be updated because they will have access to new applications and will lose access to some of the ones they had before (those specific to their group). This may involve changing their groups and the information they have access to.

4) The working environment is fairly consistent across the organisation, but varies depending on the applications allowed to be use in the group. Moreover, the application suite common to everyone is the same, which gives the user a strong sense of consistency.

The **drawbacks** of this design are as follows:

1) Now the idea of a work group exists and is very useful for sharing information within the group and for working on specific applications in a coordinated manner. The user is aware that they belong to several work groups (when applicable) and therefore sees applications and information that their co-workers should not necessarily see.

2) Now, a user's directory can be shared to people in their work group, if desired. If so, it will depend on the groups and privileges available, as it may be open to all their work groups.

The **design** should reflect the organisational structure. In contrast, the design conditions how the IT system works, as it defines it.

Therefore, careful consideration must be given to the design to be adopted, considering the groups that will exist within the organisation, their permissions for applications, and the individuals who should belong to each group. This can be done using a permissions table such as the one shown in Table 2, which lists the permissions that each user has for each application.

Reflection

Is the design different in your organisation? Leave a comment in the course forum.

Table 2. User permissions table

Group	Person	Person
Application	Permission	Permission
Application	Permission	Permission

Next, the applications/groups table (see Table 3) should be completed, listing all applications and groups. By including all the organisation's applications, this table provides an overview of all the software in use. This is particularly relevant for software that uses shared information or information accessed from databases.

Table 3. Group permissions table

Software	Group	Group
Application	Permission	Permission
Application	Permission	Permission

For example, if a study were conducted for a public organisation such as a hospital, a sample of tables might look like this:

Table 4. Permissions table example

Doctors	Joan	Carme
Appointments	R/W	R/W
Prescriptions	R/W	R/W

Administration	Maria	Pere
Accounting	R/W	R/W
Billing	R/W	R/W

Administration	Maria	Pere
Appointments	R/W	R/W
Prescriptions	R/W	R/W

Software	Doctors	Administration
Accounting	R/W	R/W
Billing	R/W	R/W
Appointments	R/W	R/W
Prescriptions	R/W	R/W
Office suite	R/W	R/W

1.5. Application distribution

The tables above provide a list of the applications that users need. The next decision to make is where these applications should be located. They can only be in two places:

1) Local (on the workstation): the application will be installed on each workstation, which will not need to search for the program on the server. This takes up more disk space on the workstation, but puts less strain on the network and is faster to run.

2) Remote: the application is installed on a server. The workstation sends requests to the server regarding the application. There are many possible variants. For example, the program may be remote (on the server) but run locally or there may only be a small client (a browser, for example) and therefore only requests for what is needed are made to the servers, with all control being handled by the server. A terminal emulation tool may also be used to connect to a host, etc.

Depending on the application in question, the decision might already be made. It might be specified by the software manufacturer, or it might be obvious that the database needs to run on a database server, meaning that the application basically must run remotely.

Generally, there are two main factors to consider when deciding where to place the application and the information it handles, and both can be local or remote. The possibilities are shown in the following table:

Table 5. Data and application location table

		Information	
		Local	Remote
Application	Local		
	Remote		
Application	Local		
	Remote		

Table 5 shows that each row "Application" has four possibilities, which only one is the best for each application installed in the organisation.

This table is populated with all the applications in the organisation. The applications are placed in the first column. It is necessary to decide which applications should be installed on each workstation, and therefore, which applications will be installed locally and which will be installed remotely.

Deciding whether to store information locally or remotely depends mainly on how many people will access it, how critical the information is, and the possibility and frequency of backups.

The list of applications used, and knowing whether they are local or remote, is critical to the design.

1.6. Application table

Following all the design steps explained so far, there are several small, scattered tables. In practice, a table such as Table 6 is needed, which summarises more than one table and can be used to extract all the necessary information.

Table 6. Global table

	Application		Information		Group	Group	Group
	Local	Remote	Local	Remote			
Application							
Application					Permission		

Permission can be R, W, or X (or a combination), which indicate read, write, or execute, respectively.

The following information can be extracted from this table:

1) The complete list of software used in the organisation. This is in the first column of the table.

2) The location of the information for each. Essentially, whether it is local or remote, i.e. whether it is on servers or spread across workstations. This is used for backup systems and to set permissions.

3) The list of user groups that must be created on the servers.

4) The list of software used by groups can also be obtained (and if the groups represent departments, etc., it can also be broken down by areas of the organisation) with the necessary permissions.

5) The list of candidate applications for configuring the base image for workstations, as well as the applications to be installed on the servers for use by users. This can be extracted from the applications that are installed remotely or locally.

All this information also gives us the starting point for the server-side design.

This table shows how it would look with the data from the previous example:

Table 7. Permissions table

	Application		**Information**		**Doctors**	**Administration**
	Local	**Remote**	**Local**	**Remote**		
Accounting	X			X		R/W
Billing	X			X		R/W
Appointments		X		X	R/W	R/W
Prescriptions	X		X		R/W	R/W

This table shows the applications that must be installed locally, i.e. on the user's machine; the applications that must be installed remotely, where the information used by the applications will reside (local or remote); which workgroups they will be part of; and what permissions they will have. Finally, there is just one thing left to do: prepare the user's workstation operating system.

The **application table** provides much of the information needed to configure the user environment on the server and clients.

1.7. The workstation operating system

Workstation operating systems are currently designed to work on a network (in corporate environments) and provide users with a graphical user interface that makes the computer as easy to use as possible. However, they are complex to install and configure, highly flexible and, unfortunately, easily misconfigured by inexperienced users. These users often complicate the system administrator's job because this operating systems' high level of flexibility also means that novice users typically feel lost when faced with IT equipment. All **network operating systems** have certain features in common that deserve consideration:

See also

See the section on "User training" to learn how to avoid these problems as much as possible.

- Users must identify themselves. Proper identification allows them to access network resources, depending on the user, as there are privileges for user groups, and to access their private information (personal data, email, etc.).

- The environment should be configured to be as homogeneous and user-friendly as possible.

- They should have quick and easy access to the most frequently used applications.

- If information is lost, the IT department should be able to restore it.

- If there is a problem with a workstation, users should know who to contact to have it resolved as quickly as possible.

The operating system must be able to communicate with the various servers (remember that these may be diverse, have different versions, and even be from different manufacturers).

2. Server design

At this point, the basic guidelines for the workstation design have been established. This includes the application locations, permissions, groups, and a general overview of the server structure.

The following section describes how this design is transferred to the servers. This design may affect the servers in the following ways:

- Number and capacity of disks.
- Content and number of disk partitions.
- Information layout on the servers.
- Number of servers.

Needs-based changes

It is not uncommon for an organisation to discover, when analysing user needs, that a complex database is required. This, in turn, leads to the need for a database server, which in turn leads to the need for a server computer, a database server and database client software on all workstations. If such things can be foreseen before the need arises it will save a lot of headaches, problems, time, user complaints and the ever-present feeling that IT never works properly.

A possible **procedure** to detect these issues could be:

1) Make a list of all the applications to be installed.

2) Check where the information for all these applications will be stored.

3) Check what permissions all these applications will need to run.

4) Based on the number of current and expected users, figure out the disk requirements. Essentially, the partitions for user data and user mail data (i.e., user mailboxes).

5) Figure out the disk requirements, considering the information that is handled and the expected amount of information to be processed.

6) Figure out the server and network requirements, considering all of the above.

Finally, the entire infrastructure must be adapted according to the findings, and changes must be made and servers expanded or replaced if necessary.

See also

Remember that these three things have already been done in the "Designing the user environment" section.

Actual user needs must be **anticipated** so that they can be reflected in the server IT structure.

2.1. Disk distribution

The general design has been outlined in previous sections, which means that the user requirements for the workstations have already been identified: the software that needs to be installed, the applications that need to run, the permissions, depending on the user and the group they belong to, and where the data will be stored (on the server, on the client, in a database). Once this design is in place, the detailed design of the server distribution can begin.

The basic partition distribution for any server is as follows:

- System partition.
- User partition.
- Application partition.
- Data partition.

See also

See the *Server administration* module.

Using the application table, the applications that users need are identified. Therefore, it is possible to know whether they are sufficient or additional ones are needed. It is also useful to find out if any application requires a server of its own.

If the organisation requires a website to publish information on the internet, a standard web server (an application) is needed on a server, and the data (the entire website) on a server (usually the same one). Everything will most likely be on different partitions (this must be decided), and whether this website will access information (databases) from the organisation must be determined to decide security issues or even whether it should be placed on a corporate server separated from the server.

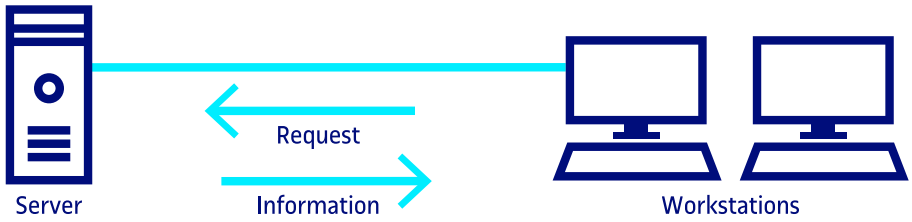
Therefore, instead of distributing information across partitions, it may be necessary to distribute it across disks within the same server.

The list of applications, the need for information, the potential number of users and their level of concurrency determine the disks distribution.

2.2. Access to information

Information from servers is delivered to users via requests over the network. As shown in Figure 6, the user makes a request to the server from their workstation, which is sent over the network. Once the request reaches the server and is processed, the requested information is returned.

Figure 6. Information access diagram



See also

See the *Server administration* module, which discusses how to optimise this problem with hardware.

The physical structure of the server disk forces that one piece of information can only be served at a time, so the multiple read requests made to the disk are placed in a queue. This issue can become severe and slow down server performance.

Detecting the problem

These problems are hard to spot because they are usually linked to network issues, as both the system administrator and users feel that requests from workstations are taking longer than expected. It is extremely difficult to detect that too many requests are actually being made to the disk.

From a design perspective, this problem can be avoided by distributing the request load across different disks or controllers (depending on the technology used), making parallel requests and, if possible, avoiding queues of stopped or collapsed requests. If the issue is critical, redundant server solutions may need to be considered.

Once again, using the list of applications that should be on the server, it is necessary to analyse how many concurrent users each one will have to assess the load.

The same must be done with the server information. If there are applications or information with a large volume of concurrent accesses, they are suitable candidates for moving to another disk or even another disk controller. If the number of requests could become critical, then they should be moved to another independent server.

Intranet web servers

Intranet web servers that access the organisation's databases are one such case. Disk loads must be monitored carefully. One of the first solutions is to use the fastest technology available (SCSI technology if considerable capacity is required or SSD if space requirements are not very high) to prevent the disk from bottlenecking the system. RAID solutions and redundant servers should also be considered.

Access to information can become critical if the number of simultaneous users attempting to access a server or disk are not carefully considered.

2.2.1. Privileges

The file system manages the process to store information on disks. This structure consists of directories, files, and an access control list for each item. Each operating system manages its file system differently, but this does not mean that information from one file system cannot be accessed by another operating system.

The file system where the information is installed must follow the user-designed structure. In other words, if user groups and permissions exist for applications, they must be included in the file systems as well as in the client and server operating systems. This provides additional system security because it is not only part of the operating system, but is also "built into" the file system itself.

3. Workstation configuration

At this stage, the applications that users need has been established, along with the permissions they require and which applications will run on the server and which on the workstation. The problem to be solved now is deciding where to store the applications chosen to run on the workstation. They can be stored on the server (they will appear as a shared drive, for example) or installed on each workstation.[4]

(4)This is explained below.

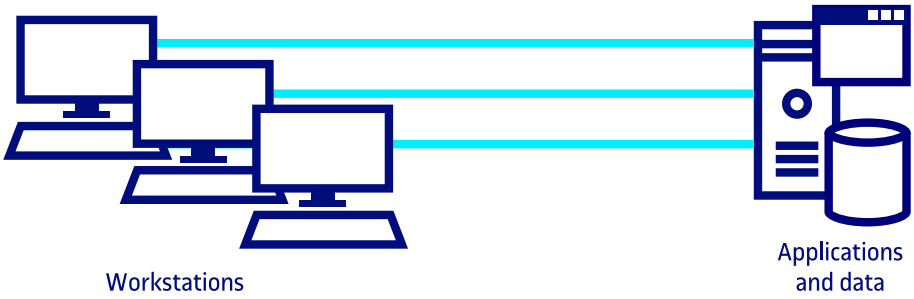
Making these decisions should be guided by the basic principle of keeping workstation maintenance as simple as possible.

A software change or update in companies with hundreds or thousands of computers can entail many days of work to reconfigure all the workstations.

3.1. Common server applications

The fewer items stored on the user's disk, the lower the risk of data loss and the time needed to restore the computer.

Figure 7. Server storage prevents data loss



Therefore, this concept is often seen as a return to the architecture of the mainframe era of the 1960s, 70s, and 80s, where users accessed a powerful central computer through simple, "dumb" terminals that did little processing and constantly communicate with servers to access programs and data, as shown in Figure 7.

See also

See the *Data management* module for some criteria for deciding where to store information.

This approach seeks to put everything on servers to avoid losing information (since it is now on all servers) and reducing the time to initially configure a workstation, since it only has the operating system because all applications are hosted on servers.

This structure presents many **problems**, including the following:

- Servers crash.
- The network crashes.
- The overall system is slow.
- Users complain about the overall performance of the system.

However, it does have some **advantages**, which are as follows:

1) There is a low risk when a workstation crashes since no data is stored on it.

2) All control is performed on the server side and, therefore, there is a low risk for disasters on client side.

3) There are also no security issues if everything is on the servers.

4) All information is on the servers.

However, this structure is not used in practice, as the disadvantages far outweigh the advantages. However, the idea is suitable for installing specific software that may be required. This software is installed on the server and run remotely on the clients. Based on network and server loads and runtime needs, it is used as a one-off solution rather than a generalised solution.

3.2. Common applications for clients

The following **criteria** are applied:

- All workstations have the same content on their hard drives (this simplifies installation).

- All workstations contain the basic software, including the operating system, office suites and software used by the entire organisation.

Designing with these criteria has many **advantages**. Here are some of them:

1) It significantly reduces network traffic.

2) It greatly increases the speed at which workstation software runs, since most applications now run locally.

3) It significantly improves overall computer performance.

4) The server only stores data and specific programs (the latter if necessary).

Therefore, users will not feel that the network is slow because many of the applications and tools will run on the workstation without requesting anything from the server. A decision must be made whether data will be stored locally (on the hard drive) or on the network (on a shared drive or in a private user space on the server).

System administrator space

A part of the server disk, which should not be visible to users, should be reserved exclusively for the administrator. This part of the disk will be used to restore workstations in case of failure and when reinstallations are necessary. Workstation recovery is a task of user administration that must be planned for. It happens almost every day when there are many workstations.

The client disks contain the software that is installed on all workstations.

3.3. Creation of the base image for workstations

Once the applications to be installed locally and remotely have been established, the application table should now be complete.

It is now time to proceed with the creation of the base image for workstations to be used in the organisation. In general, the **procedure** is as follows:

1) Operating system installation.

2) Application installation.

3) Install the clients for the applications that run remotely.

4) Configure all the operating system options to match the needs of the organisation.

5) Run tests for a while.

Once the tests have been carried out with all user groups, privileges, applications, etc., and the workstation is working correctly, the base image is complete.

The **base image** represents the software design and configuration that all the computers in the organisation should have.

It must be carefully designed, considering many factors, such as:

- **User environment**. What will the user see when they boot the computer? What will it ask them, what windows and colours will it have? What can be modified in the environment?

- **Network**. How will the network be identified? What can be done on the network? What user groups will there be? What permissions will they have?

- **Software**. What applications can be made available? Which applications will be local and which will be remote? Where will email be located?

- **Ease of use**. Everything must be designed to make the user's work easier and help them quickly become familiarised with this work tool. It should improve their performance.

- **Information**. Once the administrator has decided where to store the data and in what format, the user should find this as automatic and transparent as possible so that they need not worry about the actual location of the data.

This requires designing a base image and then cloning it as many times as there are workstations in the organisation. If necessary, the cloned workstation will then be adapted to the workstation to be used.

The general steps are as follows:

1) Prepare the base image. A computer is configured exactly as the organisation wants all its workstations to be, with the software, security, disk partitions, network configuration, etc. It is thoroughly tested to make sure everything works correctly.

2) The base image hard drive is cloned using network hard drive cloning software and saved on the server. This image can typically take up several hundred MB (or even a few GB).

3) Cloning software can create a mobile client to restore a workstation from the server. This operation results in a workstation with the software, security settings, and network configuration that has been established on the base image.

4) Finally, the settings must be adjusted for the user or workstation for which it is intended.

For computer system administration, the ideal situation is for all workstations to have the same software and hardware. Hardware uniformity simplifies purchasing, repairs, maintenance and replacement of parts, as computers break down and need servicing over time. Since this is impossible in reality, it should at least be a goal.

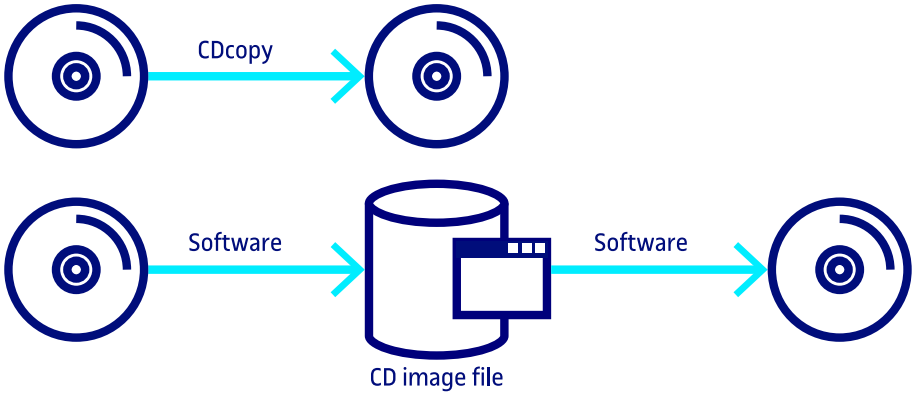
The same applies to software, which should also be standardised, but this is not possible in practice either, as this goal can never be achieved.

3.3.1. Disk images

Everyone knows about a CD[5] copying tool. This tool allows us to make an exact copy of a CD onto another CD. By doing so, if the original disk is damaged, a copy will be available, which is an exact duplicate of the original. In some ways, it is similar to photocopying, making **a faithful copy of the original**. How does it work if a duplicate of a CD is required, but there is no second CD available? As shown in Figure 8, software is available to make a copy of the CD, but instead of saving the result on a CD (the copy), it is saved in a file. This file cannot be read or written, nor can it be executed as a CD. Once the second CD is available, transfer the file to it and you will have the copy.

(5)Currently, CD devices are mostly misused for the size they can store. DVD and Blu-Ray are still used for personal backup in some cases. However, the cost per bit is comparable to the cost of acquiring an additional hard disk which have become less popular. In this section, all disk technologies are referred to as 'CD'.

Figure 8. CD duplication process using an image



With this software, users can make as many duplicates of their CD as required, without needing the original CD. The file created can be stored on the hard drive for as long as necessary. The file saved on the hard drive is called a **CD image**.

Although a CD image may seem obsolete and not very useful nowadays, the copying process can be used to copy an entire hard drive. An image of a hard drive can be made and stored on another device.

Hard drive image

Making an image or copy of a hard drive is also called *cloning a hard drive*.

The process of making a hard drive image results in a massive file (several GB in size) that contains the image of the copied hard drive. If the original hard drive is damaged (from a software perspective, i.e. the system is misconfigured

or degraded to the point that the drive must be reformatted or the system reinstalled), the **disk image can be restored** to the hard drive, so that in a few minutes the hard drive, and therefore the computer, is back up and running.

The only requirement for making a disk image is software that creates the image file somewhere. Of course, the disk image cannot be created on the same disk. If a disk image is to be stored on the same physical disk, it must be placed on a different partition. If the disk fails physically or mechanically, the image cannot be recovered, so this is not the most recommended solution. Some recommended locations are as follows:

- On the server, on a partition or administrative disk that is inaccessible to users.

- Generate the image file and then transfer it to an external hard drive. Then, if the disk image needs to be restored, it can be done from any CD reader.

- In a dedicated user administration account.

- On an external magnetic tape device on the server (DAT, DLT, AIT, etc.).

- On external backup devices such as USB drives or similar.

Like backups, when disk images are restored, the state of the disk is the same as when they were made, so any existing data may be lost. In such cases, backups should be used to recover the information. For that reason, disk images can be used for making base images for workstations.

To perform these tasks, software known as image managers can be used. Some examples are OpenGnsys, IBM Tivoli Provisioning Manager, FOG Project, Rembo Wizard & Rembo Toolkit. These tools provide environments for image generation, management and deployment.

4. Workstation maintenance

This section discusses the steps taken to keep the hardware and software in workstations running smoothly.

Workstation maintenance refers to all actions necessary to ensure that the device works properly for the end user.

In practice, this is divided into two distinct parts. The first is user device maintenance (both hardware and software), and the second is the tasks necessary for the system to function properly.

4.1. Device maintenance

Maintenance, replacement, or updating of workstation hardware and software is an important part of user administration. The maintenance tackles different tasks:

- Repairing serious hardware faults that require significant parts replacement. Reinstalling computer software is sometimes necessary.
- Replacing the computer with a new one due to hardware upgrade needs.
- Installing hardware as part of an upgrade or modernisation plan.
- Assigning the workstation to a new user and, as a precaution, destroying the information and reinstalling the user software.
- Reinstalling the computer after it has been moved to another department within the organisation.
- Reinstalling the computer due to a new role within the organisational structure.

In most cases, the IT manager is aware and the organisation's strategic plan is key to carrying out these actions within the organisation's global plan. However, other situations such as the following may also arise:

- Complete misconfiguration of the machine due to viruses.
- Disk deletion due to viruses.

- Partial disk deletion due to accidental user misuse.
- Power failure causing computer misconfiguration.
- Power failure causing a hardware problem that requires the software to be reinstalled.

There are many situations that can cause a computer to malfunction. Although it is important to install and configure the workstation for the user, it is also fundamental to establish how to recover a computer in the event of a disaster that leaves it inoperable so that it can be up and running again as soon as possible. A good maintanance model ensures proper recovery. For instance, the disk image/cloning method is perfectly applicable to computer maintenance. It can restore a computer in a very short time, assuming that the issues are only software-related. If the problems are hardware-related, a few replacement parts should be kept on hand for frequent and easy-to-repair faults (when providing a hardware repair service).

Reflection

Are there any other situations that might lead to reinstalling computers? Leave a comment in the course forum.

There are many reasons why a workstation might become inoperable. It is important to be prepared for the most common ones.

4.2. Extracting data from a computer

Many potential problems can arise. A lot of them can be solved without having to physically go to the computer. However, one scenario is particularly challenging. Although policies might be in place to ensure that data is stored on servers and that no information is stored on workstations, this is often not the case. With a corrupted operating system (which forces cloning to restore the computer to its original state), sometimes it is not possible to connect to the network to copy the information, to extract the information stored on the hard drive. In this case, finding a way to copy it is essential. The computer may not even boot up, by using a CD or memory stick to boot up, the computer can be started up and the information accessed. What is the best way to extract it, when knowing that it is in good condition? There are two ways to do this:

1) Move the hard drive. The idea is simple: remove the hard drive from the computer and put it in another computer that works. Install it as a non-primary drive and start the computer. The operating system should detect this other hard drive and all the files should be visible. The important files are copied to the main hard drive, and then the cloning operation can be performed on this hard drive (which will destroy its contents).

2) Use an external hard drive or linear tape-open (LTO). If the servers have been properly designed, the user disks will contain little information. This makes it easy to back up data using small, high-capacity mass storage devices.

The procedure is as follows: one of these devices (an external hard drive, for example) is connected to the USB port. The computer boots up with a memory stick and, from that same stick, the device is recognised. Therefore, you have a unit configured as another hard drive. Once the copy has been made, you can proceed to repair the computer. Meanwhile, for example, this information can be placed in the user space on the server. By doing this, when the problem is solved, the information will be available and can be returned to its proper place.

This requires time and money to acquire these additional tools to maintain and recover data as quickly as possible with minimal loss of information.

Confronted with a computer that has information and is not working requires investigating how to extract this important data.

4.3. Periodic maintenance tasks

Maintenance tasks are critical for the overall system to operate properly, but they are not a response to any extraordinary situation. They must be performed periodically, and most are transparent to the user.

4.3.1. Server maintenance

As mentioned above, there are numerous tasks that fall into a grey area. Is the user administrator or server administrator responsible for these tasks? They affect the server, but they also directly affect the user. Some examples of these tasks include:

- **Ensuring that users' mailboxes do not become full**. Generally, a script is used to monitor the space in users' mailboxes and, if any of them become excessively full and before the mail server crashes due to lack of space, a notification is sent to the user to clean up their emails. This task must be carried out periodically.

- **Checking that user directories do not become full**. Following the same criteria as above, it is necessary to avoid running out of space on the user partition. The size of this partition should be monitored periodically. It is also necessary to notify users of directories that exceed a certain size so that they can clean them up.

Reflection

Are there any other tasks? Leave a comment in the course forum.

4.3.2. Viruses

Viruses are one of the major problems that all user administrators face. Nowadays, antivirus software operates centrally, i.e., the antivirus is installed on a computer that acts as a "server", and the computers, users, permissions, etc. are defined. When the user logs into the system, the antivirus software is automatically installed on the computer. The administrator updates the **antivirus signature** file daily, which is automatically updated on all computers in the organisation when the user logs in. The software is also updated periodically, following the same procedure for the entire organisation.

See also

See the *Security administration* module for more information about viruses.

When running, the antivirus programme monitors all information entering the workstation (especially via the Internet) and email, and constantly scans the workstation's local disks for viruses. What happens if it finds a virus?

1) It can remove it.

2) It cannot remove it. In this case, the user can try to delete the file. If the file is critical to the operating system (often unknown to the user), deleting it may endanger the system's ability to operate. In such cases, the best course of action is always to notify the user administrator.

Whatever the case, if a virus is detected or suspected on your computer, it is a good idea to notify the user administrator by phone, as they are aware of the danger and spread of the virus. It may not be enough to simply remove the virus with antivirus software and say nothing. The virus may already have spread throughout the organisation (or worse, outside it).

This is the message that must be communicated to users to avoid spreading the virus. It is critical to train users in the use of IT tools.

4.3.3. Remote control

Remote control is a client/server technology-based software that provides access to a physically distant computer and its data via the network, manages the system and assists users with any problems they may encounter.

See also

See the section on "Help desk".

Based on client/server technology, this software has its server part on the user's workstation dedicated to serving commands issued from the client station located on the user administrator's workstation.

Remote control software is extremely useful and has added new features such as searching for items on the network, self-installation on server stations, shared connections to workstations, easy data transfer and many others.

There are many **advantages** to using remote control:

1) Cost-effective. By reducing staff, time and travel, a return on investment is guaranteed.

2) Remote working. It makes it possible to work remotely, making specific tasks more flexible, for example at weekends via working from home.

3) Fast and efficient assistance. User support is greatly improved, as it allows technicians to access the system and check existing problems for themselves. This in turn means that issues can be solved without the need for travel.

4) Training. It allows remote training through a shared connection to a workstation.

5) Maintenance. Substantial improvement in workstation maintenance.

However, some aspects can make the administrator's tasks more difficult:

1) Security. Information between the client station (server) and the administrator station (client) is transmitted over the network. If this is not secure, it could compromise control.

2) Network resources. They consume significant bandwidth, as the workstation screens travel over the network. Remote control software includes built-in tools that allow you to choose the screen resolution and colour to avoid excessive consumption.

3) Specific communication. Communication is established through ports that are often not visible from remote networks due to network features that prevent communication, usually for security reasons.

4) Legal issues. Critical. Legal regulations may be breached if the user is not notified of the connection to their workstation. Their right to privacy may be violated.

This software is designed to improve the service provided to users.

4.3.4. Deferred update

When installations are geographically dispersed or there are numerous workstations, what is the best way to update the software on those workstations?

Specific software is available that can do this. It can select the software and the target workstations and then proceed with the mass update, without having to physically move to each computer or perform the operation repeatedly.

4.4. Documentation and procedures

Documentation and procedures are often overlooked. Procedures are a technical documentation issue for system administrators.

4.4.1. Procedures

Given the high number of users, many tasks tend to be repetitive. This typically makes it useful to describe the steps involved in performing a task. Sometimes a task consists of many steps and, even though it is performed frequently, it is carried out at intervals that are long enough for it to be forgotten. This set of steps to perform a task is called a *procedure* and, to make it easy to carry out when needed, it is written down, i.e. documented.

The formal (algorithmic) definition of a procedure is as follows: an unambiguous and precise description of actions to be carried out to solve a well-defined problem in a finite amount of time.

An **action** is a finite event in time that has a defined and expected effect.

A **process** is the execution of one or more actions.

It can be defined as follows to suit the current situation:

A **procedure** is a description of the set of actions to be performed to complete a specific task.

All procedures should be recorded in a document.

Procedure document

The document may be in paper, HTML or any other format.

Each procedure should therefore be written in a document. Whenever any task needs to be performed, this *procedure* will simply need to be checked and the actions specified to carry out the task will be performed.

There are many ways to collect this information. One is in the form of an FAQ (frequently asked questions). This is posted in HTML format on a web server so that technical staff can check it at any time and from anywhere. Often, files are attached to the documents because the web format makes it possible to transfer files at the same time.

Procedure examples

The following are examples of procedures:

- Register a user.
- Set up a printer.
- Set up a workstation.
- Set up email.
- Restoring an image on a workstation.

Reflection

Is there any way to document standard procedures in your organisation? Can you think of any other ways of doing this? Leave a comment in the course forum.

None of these tasks can be performed with a single action. Sometimes a procedure can involve actions on both the server and the workstation.

4.4.2. Software

Similarly, the user administrator should ensure that users have access to some form of documentation (in any format) on how to use all available software. This makes it easier for users to learn about the tools they use. Documentation should ideally be available in several formats. It should always be easily accessible to users. Tutorials for the software are also very useful. Many software programs include tutorials, but others can be found for free on the internet. It is important to provide users with some kind of documentation on the tools they use. This shows them that administrators care about them. This is not intended to replace training, but rather to complement it.

5. User training

One aspect that is often overlooked in organisations is the user **training plan**. This should be led by the IT manager according to the organisation's guidelines and strategic plan.

Reflection

What are your thoughts on user training? What do you think of self-learning courses? Share your thoughts in the course forum.

The **advantages** of a training plan can be summarised as follows:

1) Improved software usage.
2) Increased staff effectiveness and efficiency.
3) Fewer incidents in the IT department.
4) Staff satisfaction.
5) Lower IT department costs.

Some **indirect results** include:

- Detection of new IT needs within the organisation.
- Increase in the amount of information in IT systems. This enables new methods of data retrieval for decision-making at management level.

Nobody is born an expert

Just as driving a car requires training (and, ultimately, a licence recognising the knowledge acquired), using a mobile phone or washing machine also requires an explanation of how they work, i.e. in a simple, educational way without technical jargon.

Many organisations feel training is a waste of time, but not doing it leads to the following problems:

1) Time wasted by users who are faced with new software or hardware without any knowledge, resulting in a steep learning curve.

2) The likelihood of errors during this self-learning phase is very significant, with unpredictable consequences in terms of time and cost to resolve them.

3) The chances of unintentional errors causing problems, failures, malfunctions, misconfigurations, etc. in equipment or servers are high, with the time and cost to IT department staff that this can entail.

4) The potential increment part of the IT department's workload (even to the point of collapse) due to indirect staff training issues must be considered.

5) Confusion, complaints, and the feeling that the software has been poorly installed or that the hardware is incorrect or defective and does not meet the real needs of the organisation are very likely to arise. This could jeopardise the IT or upgrade plans that are being implemented.

None of this means that the training plan should always be the same and general for everyone. A good training plan is carefully designed and considers what is explained and to which group it is explained.

The same application generates different training courses for different groups.

An integrated payroll application will have a different training course for the human resources department, for accounting, for all employees who must clock in and out, and who can check their clock-in and clock-out records on the intranet, etc. Lastly, the training course for the IT department on this application should also be different.

Therefore, there are several training plans:

1) Training plans for updates

Training plans for updates are limited to simply updating users on software or hardware changes. They are brief and help staff remain up to date. They avoid many problems in the IT department.

Training sessions are very short, which is why they are often called *sessions*, *seminars*, *workshops*, *talks*, etc. They have another significant secondary function, which is to maintain the image of the IT department as being concerned about users.

2) Training plans for new software implementation

These are the most complex, as they usually require preliminary sessions to listen to the ideas, suggestions and proposals of the group involved, and then integrate all of this into the software that is being implemented. The most challenging part of these sessions is that implementing new systems involves changing processes and ways of working, and this is generally difficult for user groups to accept. It is best to discuss this at this stage because:

Implementing software

Implementing software can mean developing a new application within the organisation itself, outsourcing it or installing standard (configurable) software.

- Users feel involved in the project, which makes them more likely to accept it.
- It is easier to overcome initial resistance by explaining their advantages with the new system (some users are still working in the usual way) and telling them that their suggestions will be considered during project development.
- The disadvantages of the current working method can be discussed by comparing it with the previous method, enabling users to notice the differences.

- Since users are involved in designing the system to make it agile, useful, and convenient rather than having it imposed, any perception of basic errors is prevented.

- Opinions, proposals, ideas, complaints, etc. should be sought on how the system should work. However, it must also be made clear that not everything that is requested is always possible and that, therefore, not all requests can be carried out.

Once the software has been developed, pilot training should be carried out with a representative group. This should serve to adjust the training plan and to detect and correct any anomalies in the installation and configuration procedures.

If pilot results are satisfactory, users can then be trained and the software installed. As soon as they find it installed on their workstations, they will be able to start using it smoothly and without causing incidents due to unfamiliarity with the software.

3) Training plans for new users

The training plan for new users should be a course with a strong focus on standard procedures, as it basically involves teaching users all the common operations used in the organisation. This avoids numerous errors and saves a lot of time, as it familiarises users with the working environment. If possible, there should be a specific focus on the tools and databases they will use in their job.

It is essential for IT department managers and company owners and managers to keep usability problems encountered by users due to software changes or updates to a minimum. Ongoing training courses for employees are crucial.

6. Help desk

Most companies nowadays have IT services that their employees need to interact with to some extent. These employees are simply users of a technology, and they do not need to know the basic principles behind it. IT services are an essential tool for these employees often, and they need to be available as much as possible.

What happens when one of these services or tools does not work properly, or not at all? The user detects a problem or incident in one of their tools, but does not have a clear idea (or technical explanation) of what is happening.

For the user, the computer is a tool to improve their self-organisation or efficiency, and they do not need to know the technical details of their device.

The same way you use a photocopier, a car, a lift or an ATM without knowing how they work internally, from the user's perspective, this should be the goal of a good system.

This is where the concept of help desk comes in to answer the following question. If the user detects a problem with a tool and does not have the necessary knowledge to solve it, what should they do?

When a user has a computer problem, they should go to a single point to resolve it: the **help desk**.

A help desk is a one-stop service that provides solutions to incidents and addresses IT-related requirements, such as computers, peripherals, IT resources, software, and platforms used by most organisations.

The help desk plays an important role in IT service delivery. It is a single point of access for employees or users who need assistance. Without a help desk, an organisation could face losses due to inefficiency.

While there are different types of help desks, such as call centres, specialist help desks and others, this section will focus on the most common type: the **three-tier** help desk.

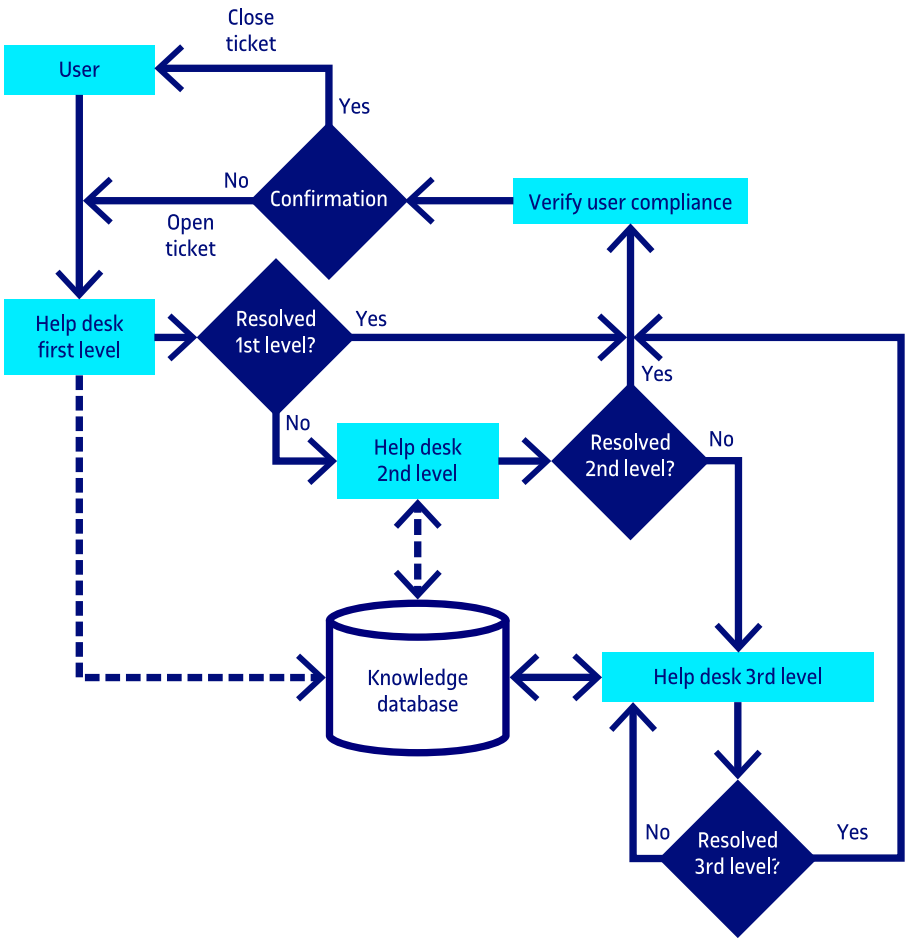
When a user detects a problem, they contact the central help desk by phone, email or other methods. At this point, their request will be received by the **first level** of support, which will record their details and the issue and open a **ticket**.

Once the ticket is open, the first level will attempt to resolve the request by checking the **knowledge base**, where they will find information and procedures for resolving incidents that have already been dealt with or are common. If the first level can resolve the request, they will verify it with the user and close it. If the request cannot be resolved, it will be passed on to the second level.

The **second level**, made up of staff with a more advanced and specialised technical profile (networks, servers, software, etc.), will attempt to resolve the request. If the assigned technician can resolve the request, they will verify it with the user and close it, also updating the knowledge base for future reference. If they cannot resolve the request, they will assign it to the **third level**.

This last level is the most specialised and often involves personnel from outside the organisation. They will resolve the request, notify the user and update the knowledge database so that this information can be used for future queries.

Figure 9. How a three-level help desk works



Each level of a help desk has specific functions assigned to it, as shown in Figure 9 and described below.

1) First level. According to the diagram shown and the help desk's range of duties, the first level of support should handle the following activities:

a) Receive incidents from users, whether by phone, email or specific software.

b) Create a request or ticket in the request management system. This is a must for proper incident management and tracking.

c) Classify the request or incident, specifying the affected group: communications, servers or others.

d) Prioritise the request or incident according to its level of severity, which may be determined by the number of users affected, the impact on production systems and other factors depending on the organisation.

e) Search for resolution information in the knowledge database to resolve your request at the first level.

f) Escalate the request to the appropriate second-level group, if necessary.

g) Update user and IT group data, if applicable.

h) Periodically check the status of open incidents and communicate with the user.

i) Prepare incident management documentation.

2) Second level. The actions relevant to this second level (specialised technicians) are as follows:

a) Reclassify the request or incident if it has been defined incorrectly.

b) Escalate the incident to the correct group if it has been assigned to the wrong group.

c) Investigate and resolve the incident. The user should be contacted if necessary.

d) Keep the knowledge base updated with information about the incident resolution, if it has been resolved. The steps taken to resolve such request must be documented correctly.

e) Develop mechanisms to prevent requests and incidents such as those that have been resolved. IT system improvement.

f) Close the request or ticket, if it has been resolved.

g) Escalate to the third level if the incident cannot be resolved.

3) Third level. At this third level, specialists in each field will be found, who will often be external to the organisation and will provide service at specific times in response to a request or problem with a very high level of resolution.

a) Reclassify the request or incident if it has been defined incorrectly.

b) Investigate and resolve the incident. The user should be contacted if necessary.

c) Keep the knowledge base updated with information about the resolution of the incident, if it has been resolved. The steps taken to resolve such request must be documented.

d) Develop mechanisms to prevent requests and incidents such as those that have been resolved. IT system improvement.

e) Close the request or ticket when it has been resolved.

6.1. Monitoring pending incidents

Someone should be assigned to check daily for any pending incidents and follow up with technicians on their status. This is usually done by someone from the first level of the help desk, as they are in contact with technicians and can check the status of incidents, update the database if necessary, and inform the user. Keeping the user informed about the status of the incident (especially if it is complex or takes a long time to resolve) is important to reassure them and give them the impression that the IT department cares about their problem.

Another very significant role of help desk support staff is to act as a filter for existing requests, beyond the role or capabilities of the IT department or the IT system implemented in the organisation (this does not mean that suggestions cannot be gathered). However, the help desk must have the ability to refuse to handle requests.

Help desk requests

A request from a user to the help desk: "I need ABC program installed on my computer." The help desk quickly realises that this software is not available in the organisation. Therefore, this request cannot be dealt with by the help desk, but as the software must be purchased, it must be channelled through the department heads, for example. This must be explained to the user and, therefore, their request cannot be dealt with.

Summary

The user represents one of the most important components of the IT system. Without users, most IT systems would be meaningless.

A well-designed environment facilitates server administration and simplifies processes. Planning ahead is key.

Similarly, good planning of workstation configurations and installations enables effective administration to resolve user problems and simplifies all subsequent processes and access. There is also a good set of tools to help with this task. Once again, planning ahead is essential.

Users must have a single point of contact in the event of IT problems. This point of contact will enable issues to be resolved quickly, queries about known situations to be answered, a response to be given within a reasonable time if the problem is one-off and, above all, users to feel that someone cares about them. This point of reference is the help desk.

Installing new software changes the way people work and must be handled carefully to avoid creating a bad working environment and to ensure that users use the new software correctly. Otherwise, they may cause the installation to fail.

User training is often overlooked, but it is key to an organisation that uses its IT system efficiently. There are different types of training depending on the user's situation. These must be considered to optimise their usefulness.

Whatever the case, users must always feel that they have the support and assistance of the IT department to carry out their tasks. This will give them the impression that the IT department is "working properly".

Activities

1. As the person responsible for user administration in your organisation, your first task is to define the access control matrix. Create this matrix for the user groups and objects that are most representative of your organisation.

2. Search the web for some remote workstation control applications that have a trial version and install one to test its usefulness. What advantages do you think it would bring to your organisation?

3. If you have access to a test workstation, create your own workstation by following the steps outlined in this module. If this is not possible, try to define the steps that would suit your organisation.

4. If your organisation has a help desk, try to define the level on which it is based. If you do not have it, try to define how you would like it to work and draw the level graph. As an additional task, you can search the internet for demo software dedicated to managing help desks to choose one that is suitable.

Self-evaluation

1. Considering the following user groups and applications for the management of a secondary school, create a table of applications and comment on the information provided.

Groups:

Students
Teachers
Management team
School administration

Applications:

Office suite
Learning tools
Assessment application (student grade system)
School accounting
School management (timetable manager, staff attendance, registration, etc.)

2. Management has requested that a time tracking system be set up within the organisation. Your IT manager meets with you, the system technicians, to provide you with the technical information needed to set up a software package to track employee clock-ins and clock-outs. This is what he does, but you remind him that a user information and training plan should be developed. The manager agrees, but asks for your opinion. What would you say?

3. A new position has been created in an organisation. A new workstation desk has been set up and you, the system administrators, have been notified of this person's arrival. Draw up a complete procedure so that when this person arrives next week, they can sit down at the new desk, start up the computer and begin working.

4. As the head of your organisation's three-tier help desk, define all the steps to be followed up to the second tier when an incident report is received from a user at a branch office approximately 200 km from headquarters.

Answer key

Self-evaluation

1. The application table is created. This is one of the possibilities.

	Application		Information		Students	Teachers	Management team	School administration
	Local	Remote	Local	Remote				
Office suite	X			X	R/W/X	R/W/X	R/W/X	R/W/X
Learning tools	X		X		R/W/X	R/W/X	R/W/X	
Assessment application		X		X		R/W/X	R/W/X	
Accounting		X		X			R/W/X	R/W/X
School management		X		X				R/W/X

The next information can be extracted:

- Complete **list of software** used in the organisation (which we already knew). This is the first column of the table.
  - Office software
  - Learning tools software
  - Assessment software
  - Accounting software
  - School management software

- **The storage location information for each application**. All the information is stored remotely, except for the data from the learning tools, as these are applications that do not update their data once they have interacted with the user.
- **List of user groups**. The list of groups shows that some users will belong to at least two groups: teachers and management team.
- **List of software used by groups**. The organisation will have the software that everyone seems to need, and each group will have the specific software for each department. Care must be taken with the information, as office automation requires detailed study. Otherwise, with permissions for everyone, all users would see all the information.
- List of candidate applications for creating the base image.
  Office suite: yes.

2. As a systems technician (or server or user administrator), it is assumed that you have been informed of an application that meets the organisation's needs, both from a technical perspective and in terms of the requirement to be met. Now, you must give your opinion on how the training plan should be developed, so you would explain the main lines of the plan.

General training

- The organisation's staff should be familiar with the time management tool.
- They should understand how they can benefit from this new tool.
- They should not see it as a control mechanism.
- List the advantages it offers. For example:
  - Clock-in and clock-out times from any computer (password protected) individualised for each person.
  - Request days off for personal matters from any computer at any time.
  - Automatic recovery of hours through the system in case of late arrival.
  - Incidents (e.g. late arrival) can be entered directly from any computer.
  - Now, this new system means that any extra time worked can be used for personal matters, which was not possible before.
  - And more.

Management training

This software requires additional training for people who manage incidents and detect time anomalies. When using another part of the software that is not used by the entire organisation, they need additional training.

Payroll training

Logically, this software must link to payroll, which is done monthly and handled by another department. Since another part of the software is used for payroll management, this requires additional training.

IT training

The application handles sensitive data, has devices for clocking in at one or more locations within the organisation, and appears to link different elements (at least payroll and administration). It is therefore a fairly complex application. Training is required for one or more people in the department to ensure that the installation is done correctly, that any problems can be resolved, and that backups are done properly.

A possible training schedule would be:

1) Initial day with the organisation's staff to assess their opinion.
2) IT training.
3) Payroll training.
4) Administration training/organisation training.

Therefore, a two-hour session would be held to explain to staff how the system works, how it is used, its advantages and the changes and improvements that will be introduced in the organisation thanks to the implementation of this software.

3. Generally speaking, the tasks may be as follows:

- Purchase a complete computer (screen, keyboard, workstation, communications card, etc.).
- Ask the appropriate department about this person's job and responsibilities to determine:
  - The physical location where the computer should be installed.
  - The group or groups to which the person belongs.
  - Therefore, which permissions they will have within the system.
  - Therefore, which software they need to use.

- Establish the physical network connection to the workstation.
- Register the user on the servers.
- Create an email account.
- Enable their private space.
- If necessary, give them special permissions for the database servers.
- Enable routers and switches, with the address on the board so that this communications board can send and receive information over the organisation's network.
- Clone the image of the workstations on this new computer.
- Adjust the configuration, given that we have the computer model: adjust the network parameters such as the computer name, IP address, configure the printers to be used, etc.
- Install specific software, if necessary.
- Test the computer.
- If everything has gone well, take the computer to the person's workstation so that they can find it when they arrive.

It is a good idea to email them explaining the basic operation (if they have not received any training from the organisation) and where they can go to solve any problems. It is also advisable to do this in writing, by telephone or in person because if they do not know how to use email, an email message will not be of much use.

4. Given that we have a three-tier help desk:

First, we will receive the incident through a single point of contact, which will be managed by the first level of the help desk. They will open an incident ticket in the help desk database and inform the user.

From the first level, the knowledge database will be accessed to try to find already certified solutions to the problem. If the issue is solved, the incident will be closed with the user. If not, it will be escalated to the second level.

At the second level, a specialised technician from the area (communications, databases, etc.) will deal with the problem and search for a solution. To carry out the appropriate actions on

the affected workstation or even to check the error on site. Remote connection tools can be used to allow quick access without having to travel.

If a solution is found, the knowledge database will be updated and the incident will be closed with the user's approval. Otherwise, it will be passed on to the third level.

Glossary

antivirus Software that searches for viruses on local or remote hard disks.

application table Summary containing the list of applications with the information and permissions associated with each group in the organisation for each application.

base image Disk image used to configure all workstations of the organisation. This is done using software, which makes the task easier.

cloning An operation that duplicates the contents of one hard disk onto another hard disk, resulting in an exact copy that cannot be distinguished from the original.
syn.: clone

core software Software deemed essential for all computers in the organisation operated by users. Typically, it includes at least the operating system, office software, a browser and an email program, as well as organisation-specific applications common to all workstations.

disk image Exact copy of a disk's contents at a given moment.

FAQ See **frequently asked questions**.

frequently asked questions A set of questions on a specific topic that Internet users repeatedly ask, which are kept on a web page along with their corresponding answers.
acronym: FAQ

help desk Part of the IT department that deals with user incidents.

ICT Information and communication technology.

incident A request from a user to solve a problem that is preventing them from working properly.

IT Information technology.

knowledge base A database with the necessary information to resolve previously resolved incidents. It can also be used to extract resolution patterns and behaviours.

office software Software that includes a spreadsheet program, a word processor, a small database, a presentation program, a calendar and, nowadays, also an email client program.

profile Information stored about the user, which, together with their ID, configures the workstation, adjusting permissions, access, graphical environment settings, etc.

remote control Control a workstation or server from a distance using a client/server program designed for this purpose.

request A request from a user to update software, install new software or configure settings.

signature file List of tags that identify viruses. Used by antivirus software to compare and find viruses.

ticket Number or identifier that tracks the request or incident opened by the user so that it can be monitored.

user environment What the user sees when they turn on the computer to work.

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